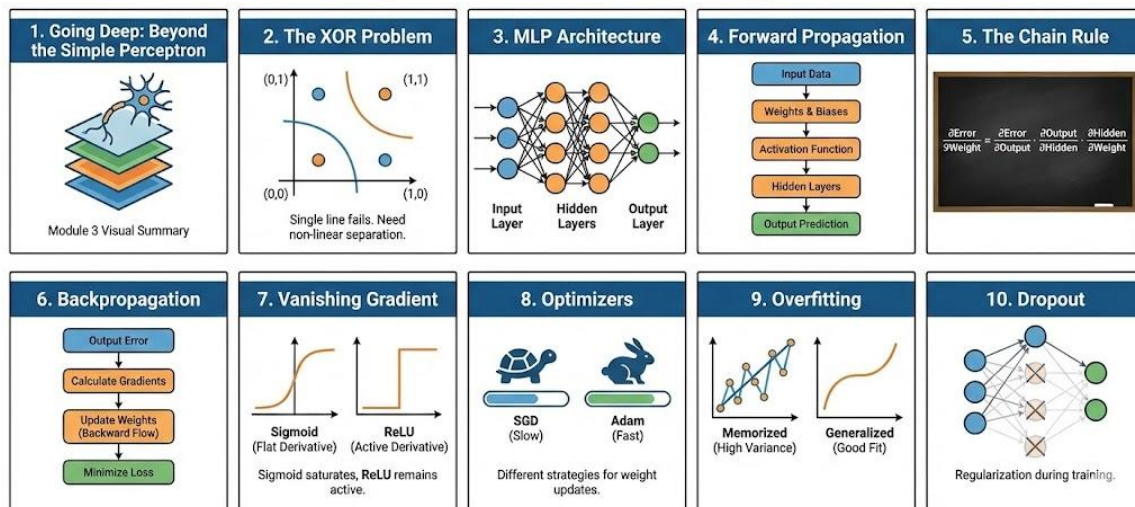


CMT 442 – Mathematical Models & Learning Algorithms

Module 3: Multilayer Networks & Deep Learning

CMT 442 Module 3: Multilayer Networks (10-Slide Visual Summary)



Module Context and Learning Objectives

Module 3 introduces deep learning as a response to the limitations of linear models. It explains how multilayer neural networks learn complex, non-linear relationships in data.

By the end of this module, students should be able to explain multilayer perceptrons, describe forward and backpropagation, identify optimization challenges, and discuss governance implications of deep learning systems.

Limitations of Single-Layer Models

Single-layer perceptrons can only solve linearly separable problems. The XOR problem demonstrates that some patterns require multiple decision boundaries.

This limitation motivated the development of multilayer networks capable of learning hierarchical representations.

Multilayer Perceptrons and Representation Learning

Multilayer perceptrons consist of input, hidden, and output layers. Hidden layers learn increasingly abstract representations of the input data.

The Universal Approximation Theorem shows that MLPs can approximate any continuous function given sufficient capacity, though practical constraints still apply.

Forward Propagation and Backpropagation

Forward propagation computes predictions by passing inputs through weighted sums and activation functions.

Backpropagation computes gradients using the chain rule of calculus to update weights based on prediction error.

Optimization Challenges

Deep networks suffer from the vanishing gradient problem, where gradients become too small for early layers to learn effectively.

ReLU activation functions and advanced optimizers such as Adam help mitigate these challenges.

Overfitting, Regularization, and Governance

Overfitting occurs when a model memorizes training data rather than learning generalizable patterns.

Regularization techniques such as dropout and early stopping help improve generalization.

ISO/IEC 42001 emphasizes documentation, reproducibility, and accountability in AI systems.

Case Study: The Accurate Model That Failed

This case study analyzes a deployed AI model that achieved high accuracy during testing but failed in production.

Students examine how overfitting, biased data, and lack of governance led to operational and regulatory consequences.

More Details

Presentation Goal: To explain how intelligence emerges from depth and hidden layers.

+1

V. Technical Deep Dive (75 Minutes)

- **Architecture Evolution:** Moving beyond the **Perceptron** (which fails the non-linear XOR problem) to **Multilayer Perceptrons (MLPs)**.

+2

- **The Learning Engine: * Forward Propagation:** Generating predictions by passing data through activation functions.

+2

- **Backpropagation:** Using the **Chain Rule** and partial derivatives to update weights based on error.

+1

- **Modern Training Fixes:**
 - **Vanishing Gradient:** Solving the "learning stop" in deep networks by using **ReLU** activation instead of Sigmoid.

+3

- **Regularization:** Using **Dropout** to prevent the model from memorizing training data (overfitting).

+1

VI. Case Study: Solving the "Black Box" Loan Predictor (45 Minutes)

- **Scenario:** A bank uses a deep MLP to predict loan defaults. While the **Training Accuracy** is 99%, the **Validation Accuracy** is only 60%, indicating **Overfitting**.

+1

- **The Problem:** The model has "memorized" the training data rather than generalizing patterns. Furthermore, under **ISO/IEC 42001**, the bank cannot explain *why* certain loans were rejected.

+3

- **The Intervention:** The data science team implements **Dropout** (disabling 20% of neurons during training) and **Early Stopping**.

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- **Governance Outcome:** The team documents the **Hyperparameters** and training data to ensure **Traceability** and reproducibility for future audits